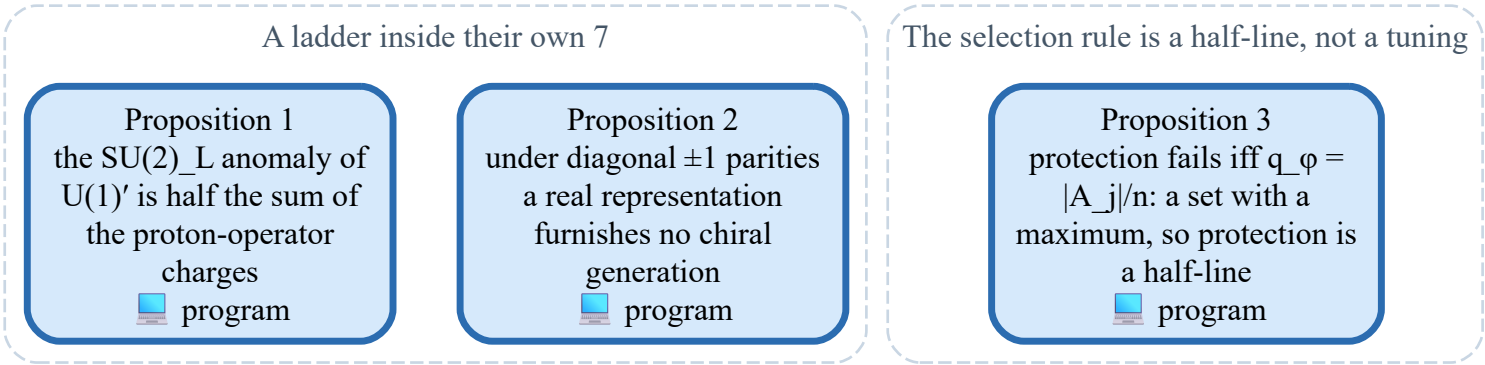
🗺️ Map of *Proton Decay in SU(7) Grand Gauge-Higgs Unification: An Obstruction, Its Minimal Escapes, and the One Row of Their Table 1 That Can Pay for Them (Part VI)*

🏷️ High-energy physics · grand gauge-Higgs unification

3 results | ✅ 0 formalized in Lean | ◐ 0 with a part in Lean | 💻 3 checked by a program | written proof only | (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



What the paper contributes

- An **obstruction** in the one-generation SU(7) model of Komori and Maru as published: three anomaly channels force $X_Q=-1/6$, where $U(1)'=T_{3L}+Y-(B-L)$ identically and the four dimension-6 $|\Delta B|=1$ operators are neutral; two further channels force a right-handed neutrino of charge -1, which only a $\mathbf{28}$ supplies (rows (2) and (3) of their Table 1).
- A **classification of the minimal escapes** inside their own group: the ladder $\mathrm{extra}(L)=(1-k)/2$, the identity $[SU(2)_L]^2U(1)'=\tfrac12\sum_jA_j$ (Proposition 1), fourteen anomaly-free three-generation assignments of which two fit their tensors; the $\mathbf{48}$ cannot host a generation because a real representation under diagonal ±1 parities is vector-like (Proposition 2, a specialisation of Gogoladze–Mimura–Nandi).
- A **benchmark selected, not a model delivered**: $V''(0)=-\pi^2\zeta(3)D$ with D rational per multiplet ($\mathbf{84}^{(+,+)}\mapsto5/4$, $\mathbf{48}^{(+,+)}\mapsto0$), so only case (2) satisfies both necessary conditions, at one-loop curvature.
- A selection rule in closed form: protection fails iff $q_\varphi=|A_j|/n$ (Proposition 3); $q_\varphi=3/2$ protects for every half-integer brane-quark charge.
- An audit of the instrument: their $\alpha_{\min}$ column is not reproduced (stated as an open question, not a defect); the verdict holds on a wedge of multiplicative repairs; two-loop safety is left open.

🚀 **Start here: Proposition 1**: the SU(2)_L anomaly of U(1)′ is half the sum of the proton-operator charges.

Main results:

- **Proposition 1**: the SU(2)_L anomaly of U(1)′ is half the sum of the proton-operator charges
- **Proposition 2**: under diagonal ±1 parities a real representation furnishes no chiral generation
- **Proposition 3**: protection fails iff q_φ = |A_j|/n: a set with a maximum, so protection is a half-line

🧭 Fields: where the paper lives and what it connects

Tools taken from other branches

- **Anomaly cancellation** → the six U(1)′ anomaly channels on the published content; the gravitational and cubic conditions as a Diophantine system (Costa–Dobrescu–Fox) — Proposition 1
- **Representation theory of SU(7)** → weights of tensor powers and of the adjoint, orbifold parities, chiral zero modes — Proposition 1, Proposition 2
- **Discrete gauge symmetry** → the residual subgroup left by a charged condensate (Krauss–Wilczek, Ibáñez–Ross) — Proposition 3

What the results serve

- **Proton stability in orbifold GUTs** ← the obstruction at one generation, the family-dependent escapes, and the selection rule for the breaking scalar — Proposition 1, Proposition 3
- **Model building in gauge-Higgs unification** ← case (2) of Komori–Maru Table 1 as the benchmark to compute next (Part VII of the series takes it up)